



天能帅福得能源股份有限公司
TIANNENG-SAFT ENERGY JOINT STOCK COMPANY

锂离子电芯规格书 Specification for Lithium-ion Rechargeable Cell

电芯型号: ITR18650-2600P

Cell Type: ITR18650-2600P

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1. 基本信息 General Information

1.1 适用范围 Scope

本规格书规定了由天能帅福得能源股份有限公司生产的圆柱型锂离子电池的技术要求，测试方法及注意事项，如需获取本规格书以外的技术要求，请与天能帅福得能源股份有限公司联系相关事宜。

This specification describes the technical requirements of Cylindrical Lithium-ion Cell supplied by TIANNENG-SAFT ENERGY JOINT STOCK COMPANY. If any other technical information is needed, please contact TIANNENG-SAFT ENERGY JOINT STOCK COMPANY.

1.2 产品分类 Product Classification

圆柱型可充电锂离子电芯

Cylindrical Rechargeable Lithium-ion Cell

1.3 型号名称 Model Name

圆柱形 ITR 18650-2600P 电池

Cylindrical ITR 18650-2600P Cell

2. 产品测试要求 Test requirement and Standard

2.1 环境要求 The requirement of environment

除非另有规定，本规格书中各项试验应在标准大气条件下进行：温度：23℃~27℃；相对湿度：45%~85%；

Unless having additional statement, All tests in this specification will be done under standard atmosphere conditions: temperature:23℃~27℃； relative humidity: 45%~85%；

大气压力：86 kPa~106 kPa

Atmospheric pressure:86 kPa~106 kPa

2.2 测试设备要求 The requirement of measuring equipment

2.2.1 电流表和电压表 Amp-meter and volt-meter

电流表和电压表的精度需达到 0.5 mA 和 0.5 mV 或者更高。

The amp-meter and volt-meter should have an accuracy of the grade 0.5 mA and 0.5mV or higher.

2.2.2 游标卡尺 Slide caliper

游标卡尺精度要求 0.01 mm。

The slide caliper should have 0.01 mm scale.

2.2.3 内阻测试仪 Impedance meter

内阻测试仪使用的频率为 AC 1KHz。

The Impedance meter with AC 1KHz should be used.

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2.3 标准充电 Standard Charging Method

“标准充电”即在标准测试条件下，电芯先以恒定电流 0.5 C 充电至 4.2 V，再以 4.2 V 的恒定电压充电至电流小于 0.02 C，搁置 1h。

“Standard Charging” means that in standard test conditions, charge the cell at a constant current of 0.5 C until the voltage reaches 4.2 V, then charge it at a constant voltage of 4.2V until the current decreases less than 0.02 C and placed for 1 h.

2.4 标准放电 Standard Discharging Method

“标准放电”即在标准测试条件下，电芯以恒定电流 1C 放电至 2.75 V。

“Standard Discharging” means that in standard test conditions, discharge the cell at a constant current of 1 C until the voltage reaches 2.75 V.

3. 标准规格 Nominal Specification

项目 Item	规格 Specification	条件 Condition/ Note
3.1 标称容量 Nominal Capacity	Typical:2600 mAh Min:2580mAh	0.2C 2.5V 放电容量 0.2C 2.5V discharge capacity
3.2 交流内阻 AC Impedance	$\leq 30 \text{ m}\Omega$	在 1000 Hz 下测量 At AC 1000 Hz
3.3 标称电压 Nominal Voltage	3.6V	
3.4 电芯尺寸 Cell Size	18.2 $\pm$ 0.3 mm	电芯直径 Cell Diameter
	65.0 $\pm$ 0.3 mm	电芯高度 Cell Height
3.5 电芯重量 Cell Weight	45 $\pm$ 1.5 g	
3.6 充电截止电压 End-of-charge Voltage	4.2 V 0.02C cut-off	恒流恒压充电 CC +CV Mode
3.8 充电方式 Charging Method	0.5 C at CC/CV	标准充电 Standard Charging
	0.7C at CC/CV	快速充电 Max Continuous Charging
3.9 放电截止电压 End-of-discharge Voltage	2.75 V	恒流放电 CC Mode
3.10 最大持续放电电流 Max continuous Discharging Current	7.8A (3C)	
3.11 最大瞬时放电电流 Max Pulse Discharging Current	13 A (5C)	
3.12 使用温度范围 Operation Temperature	0 to 45℃	充电 Charge
	-20℃ to 60℃	放电 Discharge

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3.13 存储温度 Storage Temperature	-20~60℃ -20~45℃ -20~25℃	一个月 1 month 三个月 3 months 一年 1year
3.14 外观 Appearance	无破裂、变形、电解液泄露等 Without break, distortion, leakage and so on	

4. 电性能 Electrochemical Performance

测试项目 Test Item	测试方法 Test Method	检验标准 Criteria
4.1 循环寿命 Cycle Life	电芯按 0.5 C CC/CV 充电后搁置 30 min, 然后以 1 C 恒流放电至 2.75 V 结束, 搁置 30 min, 再进行下一次循环。 Cell shall be charged at CC/CV mode(CC: 0.5 C, CV: 4.2 V, End-of-charge current: 0.02 C); After stored for 30 min, cell shall be discharged at CC mode(1C, End-of-charge voltage: 2.75 V);After stored for 30 min, continue to the next cycle. 电芯按 0.5 C CC/CV 充电后搁置 30 min, 然后以 3 C 恒流放电至 2.75 V 结束, 搁置 30 min, 再进行下一次循环。 Cell shall be charged at CC/CV mode(CC: 0.5 C, CV: 4.2 V, End-of-charge current: 0.02 C); After stored for 30 min, cell shall be discharged at CC mode(3C, End-of-charge voltage: 2.75 V);After stored for 30 min, continue to the next cycle.	容量保持率: ≥800 次@80 % Capacity retention: ≥800 cycles @80 % 容量保持率: ≥400 次@80 % Capacity retention: ≥400 cycles @80 %
4.2 倍率放电性能 High-rated Discharging Performance	电芯按 2.3 规定充电后, 室温下以 3 C 电流放电至 2.75V。 Cell shall be charged per 2.3, and discharged at 3C to 2.75V at RT.	放电容量: ≥ 95% Discharge Capacity: ≥ 95%
4.3 低温性能 Low Temperature Performance	电芯按 2.3 规定充电后, 将电芯放入一定温度的低温箱中恒温 24 h, 然后以 1 C 电流放电至 2.5V。 Cell shall be charged per 2.3 and stored in a temperature-controlled environment for 24h. Then discharged to 2.5V at 1C.	-20℃放电容量: -20℃Discharge Capacity: ≥ 70%
4.4 高温性能 High Temperature Performance	电芯按 2.3 规定充电后, 将电芯放入一定温度的高温箱中恒温 5 h, 然后以 1C 电流放电至 2.75V。 Cell shall be charged per 2.3 and stored in a temperature-controlled environment for 5h. Then discharged to 2.75V at 1C.	55℃放电容量: 55℃Discharge Capacity: ≥ 95%
4.5 常温存储 Room Temperature Storage Test	电芯按 2.3 规定充电后, 在 25 °C ±2 °C 下搁置 28 天, 再以 1C 恒流放电至终止电压。 Cell shall be charged per 2.3, then stored at 25 °C ±2 °C for 30 days. Finally discharged to ending voltage at 1 C.	容量保持率 ≥ 90 % Capacity retention ≥90% 容量恢复率 ≥ 97 % Capacity recovery≥97%
4.6 高温存储 High Temperature Storage Test	电芯按 2.3 规定充电后, 在 55 °C ±2 °C 下搁置 7 天后, 室温搁置 5h, 再以 1 C 恒流放电至终止电压。 Cell shall be charged per 2.3, then stored at 55 °C ±2 °C for 7 days. After standing for 5h, discharged to ending voltage at 1 C.	容量保持率 ≥ 90 % Capacity retention ≥90% 容量恢复率 ≥ 97 % Capacity recovery≥97%

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5. 环境特性 Environment Characteristic

测试项目 Test Item	测试方法 Test Method	检验标准 Criteria
5.1 温度循环测试 Thermal cycling Test	电芯按 2.3 规定充电后, 放入恒温箱中, 60 min 内降至 -40 °C, 保持 90 min 后, 在 60 min 内升至 25 °C, 再在 90 min 内升至 85 °C, 保持 110 min, 然后在 70 min 内降至 25 °C。重复上述步骤 5 次, 观察 1h。 Cell shall be charged per 2.3, and put into an incubator. Temperature inside the oven will drop to -40 °C in 60 min and remain for 90 min. Then it will rise to 25 °C in 60 min and keep rising to 85 °C in 90 min, following by remaining for 110 min. And it will drop to 25 °C in 70 min. Repeat this process for 5 times, then check it for 1h.	电芯不爆炸、不起火、不漏液 No explosion, no fire, no leakage.
5.2 低气压测试 Low-pressure Test	电芯按 2.3 规定充电后, 放入低气压箱中, 调节气压为 11.6 kPa, 温度为室温, 静置 6 h 后, 观察 1h。 Cell shall be charged per 2.3, then stored it for 6h at an absolute pressure of 11.6 kPa (RT). Check it for 1h.	电芯不爆炸、不起火、不漏液 No explosion, no fire, no leakage.
5.3 跌落测试 Drop Test	电芯按 2.3 规定充电后, 正负端子向下从 1.5 m 高度自由跌落到水泥地面上, 观察 1h。 Cell shall be charged per 2.3, then dropped from a height of 1.5 m onto the concrete ground. Positive and negative terminals of cells shall be towards the ground. Check it for 1h.	电芯不爆炸、不起火、不漏液 No explosion, no fire, no leakage.
5.4 浸泡测试 Soaking Test	电芯按 2.3 规定充电后, 完全浸入 3.5 wt% NaCl 溶液中 2 h, 观察 1 h。 Cell shall be charged per 2.3, then completely soaking into NaCl solution (3.5 wt %) for 2h. Check it for 1h.	电芯不爆炸、不起火 No explosion, no fire.

6. 安全特性 Safety Characteristic

测试项目 Test Item	测试方法 Test Method	检验标准 Criteria
6.1 短路 Short-Circuiting Test	电芯按 2.3 规定充电后, 将正、负极经外部短路 10 min, 外部线路电阻 5mΩ; 静置 1h。 Cell shall be charged per 2.3, then short-circuited by connecting the positive and negative terminals with a resistance of 5mΩ for 10 min. Check it for 1h.	电芯不爆炸、不起火 No explosion, no fire.
6.2 过充电 Over-charge Test	电芯按 2.3 规定充电后, 以 1C 电流充电至 6.3 V 或充电达 1h 后停止充电, 观察 1h。 Cell shall be charged per 2.3, then charged at 1C to ending voltage of 6.3 V or charged at 1C for 1h. Check it for 1h.	电芯不爆炸、不起火 No explosion, no fire.
6.3 过放电 Over-discharge Test	电芯按 2.3 规定充电后, 以 1C 电流放电 90min, 观察 1h。 Cell shall be charged per 2.3, then discharged at 1C for 90 min. Check it for 1h.	电芯不爆炸、不起火、不漏液 No explosion, no fire, no leakage.
6.4 挤压测试 Crush Test	电芯按 2.3 规定充电后, 以半径为 75 mm 半圆柱体垂直电芯极板方向, 以(5±1) mm/s 速度挤压, 当电压为	电芯不爆炸、不起火 No explosion, no fire.

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	0 V 或变形量达到 30%或挤压力达到 200 kN 后停止挤压, 观察 1h。 Cell shall be charged per 2.3, then crush the cell perpendicularly to the cell plate at a rate of (5±1) mm/s with a semi-cylinder (radius of 75 mm). When met any of the following criteria, stopping crushing and check it for 1h. 1. Voltage reaches 0 V; 2. Deformation reaches 30%; 3. Pressure reaches 200 kN.	
6.5 加热测试 (130 °C) Heating Test	电芯按 2.3 规定充电后, 放置入温度箱, 以 5 °C/min 的速率由室温升至 130 °C±2 °C, 并保持 30 min 后停止加热, 观察 1h。 Cell shall be charged per 2.3, then heated in an oven. Temperature will rise to 130 °C±2 °C at a rate of 5 °C/min and remain for 30 min. Check it for 1h.	电芯不起火, 不爆炸 No explosion, no fire.

7. 存储与运输 Storage and Transportation

7.1 由于电芯的特性, 需要对电芯进行合适的包装来保护。

Based on the character of cell, proper environment for transportation of pack need to be created to protect the battery.

7.2 运输过程应采取谨慎的预防措施以避免电芯在运输过程中损坏。

During transportation, careful precautions should be taken to avoid damage of cell.

7.3 电芯需在 0 °C-45 °C 的干燥、清洁、通风的环境下存储。

Cell should be kept at 0°C-45°C in warehouse where it's dry, clean and well-ventilated.

7.4 装卸电池时需注意避免跌落、翻转和堆积。

During loading of battery, attention must be paid against dropping, turning over and serious stacking.

8. 安全守则 Precautions and Safety Instructions

为避免电芯泄露, 过热和爆炸, 请注意以下事项:

In order to prevent the battery leakage, getting hot and explosion, please pay attention to preventing measures as following:

警告! Warning!

- 请勿将电池投入水中。非使用时, 电池需在干燥阴凉处存放。
Never throw the battery into water. Store it under dry, shady circumstance when not use.
- 请勿颠倒正负极使用。
Never misidentify the positive and negative terminals.
- 请勿直接用金属接通正负极, 避免短路。

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Never connect the positive and negative terminals of battery with metal to prevent short-circuiting.

- 请勿将电池与金属一起储存或运输。

Never ship or store the battery together with metal.

- 请勿敲击、投掷或踩踏电池。

Never knock, throw or trample the battery.

- 请勿用钉子或其它利器破坏电池。

Never cut through the battery with nail or other edge tool.

使用建议！Tips!

- 请勿在过高的温度下使用或储存电池，否则会引起电池过热，致使寿命减短和性能降低，甚至起火。

Never use or store the battery under the over-high temperature. Otherwise it will lead to battery over-heating, which might lose some function and reduce life, even getting fire.

- 请勿将电池投入火中或其它热源中，避免起火、爆炸和环境污染。废电池需回收至供应商处，移交回收站处理。

Never throw the battery into fire or heating machine to avoid fire, explosion and environment pollution; scrap battery should be returned to the supplier and handled by the recycle station.

- 请勿将电池置于强静电场或强磁场中，否则会破坏保护设备。

Never use the battery under strong static electronic and magnetic field, otherwise it will destroy the protecting device.

- 若电解液泄露并进入眼睛，请勿揉捏，经水洗后尽快就医。

Never knead eyes if leakage electrolyte gets into eyes. Wash eyes by water and seek medical advice ASAP.

- 若电池在使用、储存、充电过程中发出异味，过热，形变或其他非常规情况，请停止使用并移除设备。

If battery emit peculiar smell, over-heating, distortion or appear any unconventionality during using, storage or charging process, please stop using and take it out of the device.

- 请勿在充电中直接插拔电池，并使用规定充电设备进行充电。

Never cut the battery in socket directly, please use the stated charger when charging.

- 使用前请检查电池电压和相关连接器；若有异常请勿使用。

Check the voltage of battery and relevant connectors before using. Do not use until everything turns out to be normal.

- 在充电前请检查相关设备绝缘性、物理状态和老化情况。电池电压需高于截止电压，如有异常，需标记并勿更改现状，然后通知我们售后服务部门，待我们工作人员到场维修。

Prior to charging, fully check the insulativity, physical condition and ageing status. The pack voltage must not be less than the cut-off voltage, if not, it needs to be labeled. The user should contact our Customer Service

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Department. It can't be charged until repaired by our staff.

- 电池需在 50 %SOC 下储存，若半年未使用，需重新充电。

The battery should be stored in 50% SOC. It needs to be charged once again if out of use for as long as half a year.

- 若电极端子污染，需用干净、干燥的棉布擦拭，否则会导致接触和操作不良。

Clean the dirty electrode with a clean dry cloth if any contamination appears, otherwise poor contact or operation failure may occur.

9. 公司地址 Company Address

天能帅福得能源股份有限公司

TIANNENG-SAFT ENERGY JOINT STOCKCOMPANY

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A0		初版		
A1	4.1	增加 3C 放电循环	19.6.18	周步清

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